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An investigation of videos for abnormal behavior detection

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Abstract

The identification of abnormal behavior has several applications. There are several ways, ranging from classical to deep learning based. It may be used to monitor campuses, banks, transportation, and airports. In many circumstances, the context determines whether real-life events are common or unusual. Recent video surveillance anomaly detection systems are good enough, but they come at a significant computational cost and require particular hardware resources. When it comes to real-time anomaly detection, extra emphasis must be paid on lowering model complexity, which causes computational and memory demands. This study attempts to find real-world abnormalities in CCTV recordings, such as violence, detention, property destruction, assault, burglary, blast and theft. The impact of these abnormalities on public safety is enormous. The study also provides a low-cost algorithm for detecting crowd irregularities. The proposed model uses the convolutional neural network based on DenseNet121 as the feature extractor. Our suggested framework has an AUC of 86.63 percent on the UCF-Crime dataset and obtains new state-of-the-art performance

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1. Introduction

Surveillance recordings are becoming more common in a range of environments to monitor human activities and prevent crime. It can be helpful if someone is viewing the footage and triggering an alarm if something out of the ordinary occurs. However, these occurrences are rare, and most of the time, the individual watching the footage will not notice anything unusual. These unusual events might be looked on as anomalies since they exhibit patterns that do not match what is regarded as normal [1]. Recent research, however, suggests that anomaly detection, in particular, autonomous human behavior analysis, is still an open issue [2]. Researchers have been attempting to develop strong anomaly detection algorithms capable of automating the process of monitoring and detecting unexpected occurrences in surveillance recordings. The main difficulties that lie in the way of anomaly identification in autonomous video monitoring are discussed as. First, the definition of an anomaly varies depending on the context. Second, the various

options of what constitutes an abnormality are limitless [3]. Third, even if abnormalities are rare, obtaining strong characteristics from the data is a challenging task [1]. There are currently just a few datasets for anomaly detection accessible, the majority of which are small-scale or confined to particular scenarios, such as UCSD Peds [4], Avenue [5] datasets. These datasets are also utilised for semi-supervised anomaly detection with normal training samples at first. Only UCF-Crime [6] dataset is presently publicly accessible for the challenge under video-level scenario, to our understanding. Therefore, we have experimented on UCF-Crime dataset. Developing a framework for anomaly detection in surveillance video feeds is the major goal of the proposed approach. This is accomplished by executing the anomaly detection algorithm on frames, which enables this system to handle data almost efficiently. This paper provides a distinctive DL-based anomaly detection approach to improve safety at campuses and airports. The model's goal is to identify and classify abnormalities found in video surveillance, such as robberies, arsons, shootings, steeling, etc. To extract detailed characteristics from the video, our model makes use of DenseNet121[21]. With respect to accuracy, speed, parameter efficiency, and model comprehension, DenseNet is a potent deep neural network. Experimental findings show that the suggested model outperforms the state-of-the-art methods on the UCF-Crime dataset. The remaining part of the paper have been arranged as follows. Section 2 provides an overview of the prior efforts, while Section 3 talks about the suggested model. Section 4 evaluates the performance of the suggested model, and Section 5 concludes the paper with suggestions for future research.

2. Related Work

2.1. Traditional Methods: Considers low level features.

The early research focused on trajectory-based anomaly. Calderara et al. [7] proposed the two methods the first being the capacity to use several video cameras to follow and identify single person trajectories in a busy space, and the second is a new class of novelty detection techniques based on graph spectral analysis. Jiang et al. [8] used the method which involved evaluating video occurrences at many levels, taking into account both geographical and temporal context. Tung et al. [9] described that in three ways, the suggested technique extends the goal-based framework. With sparse reconstruction analysis, Li et al. [10] offer a novel trajectory-based visual abnormal behavior detection technique. However, because the technique strongly relies on continually monitoring the item of interest, one key problem is occlusion. Because of these disadvantages, low-level characteristics were emphasized for feature extraction instead. Zhang et al. [11] introduced a new anomaly detection framework that combines motion and appearance signals to recognize unusual objects and behaviors in video and used support vector data description. Wang et al. [12] proposed the OCELM model. These conventional procedures, despite their shown efficacy on benchmark datasets, are still ineffective when applied to a different domain. Additionally, they struggle to adjust to irregularities they have never encountered. Due to these factors, the majority of recent research has been on using Deep Neural Networks to detect anomalies. There is no longer a need for manual feature development because these neural networks produce useful and differentiating traits on their own.

2.2. Deep learning

Deep learning frameworks are ideal for improving accuracy when spotting anomalous human behavior in pictures or recordings as the deep learning methods are more reliable and accurate [26]. Waqas et al. w constraints [5] proposed the UCF-Crime dataset and the proposed the model having the bags with normal frames as negative bags and the bags with anomaly as positive bags. Then the videos are sent for extraction in the 3D ConvNet and finally the scores are generated for positive and negative bags. The model achieves an AUC of 74.44 and the model without constraints achieves an AUC of 75.41. Using the datasets from the Avenue, UCSD pedestrian, and Subway, Hasan et al. [13] trained the algorithm. They used the convolutional and pooling layers for encoding and the deconvolutional layers and un pooling layers for decoding while accounting for the size of the input cube and training data. Wang et al. [14] introduced a unique one-class learning formulation based on subspaces in a discriminative context, with these subspaces sandwiching the data. According to Zhong et al.'s [15] methodology, the action classifier generates noisy snippet-level classifications by extracting spatiotemporal information from abnormal video clips. Dubey et al. [16] made use of the abnormal and normal videos are used to train the network and the normal videos are divided into positive and abnormal videos into negative instances. 3D Resnet was used for feature extraction. Kamoona et al. [17] propose that the model deals with video clips as sequential visual data and build a temporal encoding network that

exploits spatiotemporal evolution between feature instances. The results of experiments using normal and abnormal videos demonstrate the effectiveness of their proposed solution. In order to make it work in a surveillance system, we suggested a minimally complicated deep learning-based model for anomaly detection. The chosen data frame sequence is provided to DenseNet121, which assists in feature extraction and further categorizes the behavior as normal or abnormal. To the best of our knowledge, DenseNet, a potent deep neural network with significant benefits in terms of accuracy, speed, parameter efficiency, and model comprehension [21], is not being investigated sufficiently for abnormal behavior detection. Additionally, when the quantitative findings are contrasted with those obtained using state of the art techniques, a significant improvement is seen, demonstrating the efficiency of our methodology. Table 1 discusses some of the deep learning models. Table 1. Deep learning methods

#	Method	Year	Dataset Used	Description
1	Hasan et al. [13]	2016	Avenue , UCSD pedestrian, Subway	Use of CNN autoencoders
2	Waqas et al. constraints [6]	2018	UCF-Crime, Tad dataset	Proposed the model having positive and negative bags having abnormal and normal frames
3	GODS [14]	2019	Dash-Cam-Pose, UCF Crime	Dash-Cam-Pose dataset is proposed and introduced a unique one-class learning formulation based on subspaces in a discriminative context, with these subspaces sandwiching the data.
4	C3D [15]	2019	UCF-Crime	Supervised learning task under noisy label.
5	MILR [16]	2020	UCF-Crime	Reducing the false alarm rate and using 3D ResNet to extract spatial temporal features.
6	Multiple instance-based video anomaly [17]	2020	UCF-Crime, ShanghaiTech	Build a temporal encoding network that exploits spatiotemporal evolution between feature instances.
7	MIST [18]	2021	UCF-Crime, ShanghaiTech	Effectively fine-tuning of encoders for the self-training framework.
8	WSAL [19]	2021	TAD dataset, UCF-Crime,	Focused on the location of anomalies in surveillance footage and suggested a weakly trained network for deep temporal context exploration in consecutive segments.

3. Proposed framework

3.1. Introduction

According to the literature study, anomaly detection is a difficult topic to address owing to the occurrence of many complicated circumstances. Our methodology has an accuracy of 86.63 percent in identifying anomalous behavior, and it is centred on the UCF-crime dataset. Our model utilises DenseNet121 to extract deep features from the video. On comparison with the other datasets UCF-Crime dataset is vast and the dataset has multiple classes we have used categorical cross entropy as a loss function in our DenseNet implementation.

3.2. Dataset Details

UCF-Crime Dataset: Sultani et al. [6] constructed the UCF-Crime Dataset since the prior datasets were rather modest in size (2018). Accidents, burglary, explosion, fighting, robbery, shooting, theft, shoplifting, and vandalism are among the 13 real world abnormalities in this dataset. These text inquiries were not confined to English; they were also conducted in other languages (using Google Translate). 1,900 videos total, including 950 uncut real-world surveillance recordings and 950 regular recordings, are included in the collection. The complete file has a resolution of 240x320 pixels and around 128 hours of data. 1900 films totalling 128 hours make up the UCF crime dataset. The dataset is useful for the problem of video categorization since the length of a video is on average 4 minutes, with 70% of videos being under 3 minutes. Each of the 14 classifications found in the UCF-Crime dataset are represented in Fig. 1.

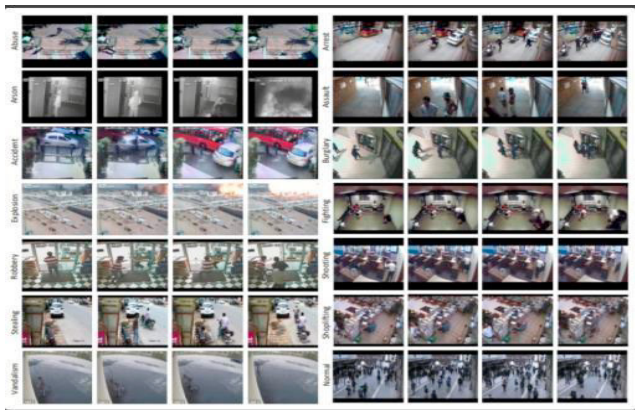


Fig.1. UCF-Crime dataset

3.3. DenseNet Model

ResNet [20] inspired DenseNet [21]. Dense connections, or the unique layer connectivity of DenseNet, play a significant role in giving specific benefits. The main advantages of such deep networks over ResNet [20] and Inception Net [22] include improved model comprehension, reduced complexity, fewer parameter requirements, lower computing and power consumption, and increased accuracy. The input data or gradient that was present in the majority of the levels by the time the last layer was reached vanishes as the network gets deeper. By directly tying together all the layers with similar feature sizes, DenseNet circumvents the gradient vanishing issue. Deeper within the network, more general features may be extracted, which is the main goal of employing DenseNet architecture as a feature extractor. The feature extraction procedure has been carried out using the 121-layer, pre-trained Densely Connected Convolutional Neural Network (DenseNet-121). The form of this model that we employed in our investigation was trained using the substantial ImageNet dataset, which is freely accessible and was developed by Huang et al. [21]. The DenseNet121 design consists of 4 dense blocks, 3 transition layers, and a convolution and pooling layer at the beginning. The categorization layer is the last layer present after these layers. Fig.2. illustrates the proposed "Densely Connected Convolutional Neural Network" (DenseNet-121)-based abnormal behavior detection.

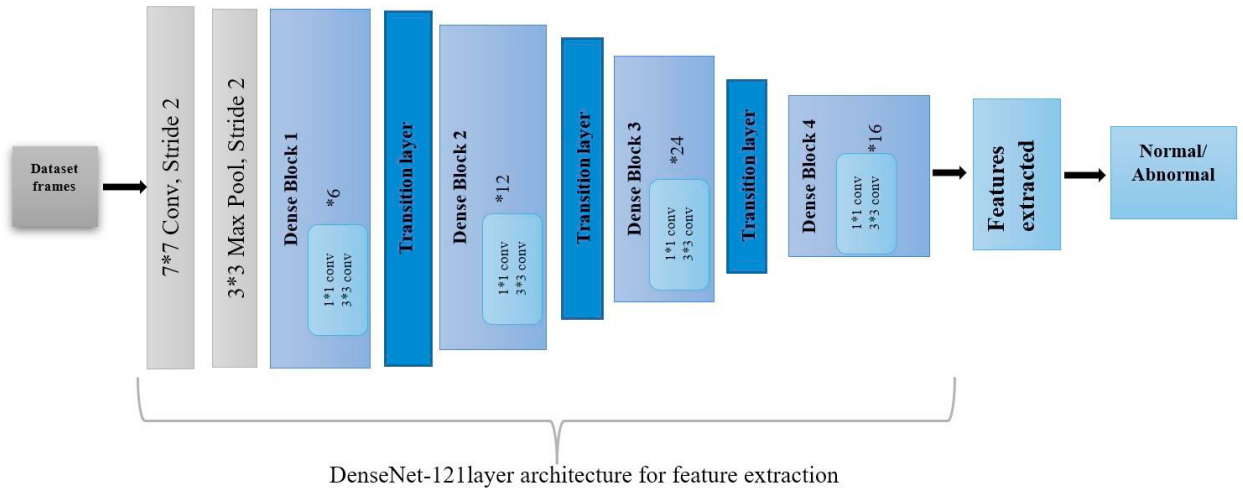


Fig. 2. Architectural diagram of the proposed model.

3.4. Training and testing data

For train and test data, our model employs two distinct generators. Fig.3. and Fig.4. show the distribution of testing and training data for all 14 classes:

'Abuse','Arrest','Arson','Assault','Burglary','Explosion','Fighting','Normal','RoadAccidents','Robbery','Shooting','Shoplifting ng', 'Stealing', 'Vandalism'. Our suggested model converges with top outcomes in terms of loss=0.111 and accuracy=0.9986 throughout the training procedure.

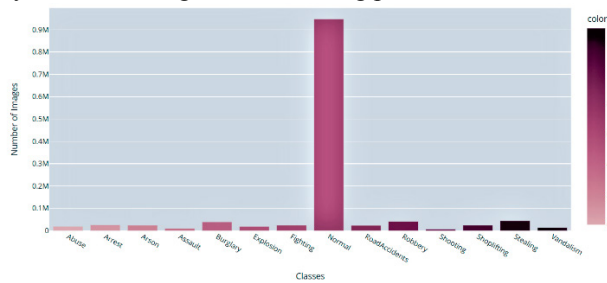


Fig.3. Train data distribution

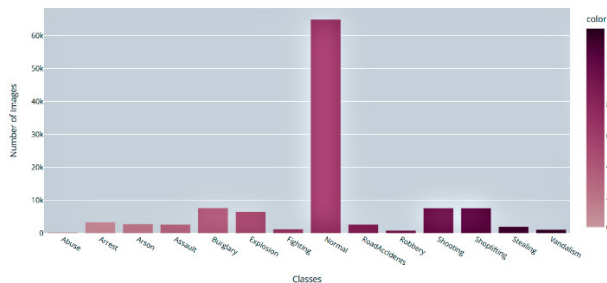


Fig.4. Test Data Distribution

4. Experimental Details

4.1. Experimental details

Here is the experimental setup:

By utilising TensorFlow upon the Keras framework with GPU accelerated performance, the network is designed to run. This made it possible to train the network using the UCF-Crime dataset in around three hours. Here in the proposed approach the model's ability to detect anomalies is assessed and contrasted with cutting-edge anomaly detection methods based on the UCF-Crime dataset. The tweaking of hyperparameters is crucial for training an accurate deep learning model. The seed, dropout, depth of network, learning rate, optimizer, and batch size are all common hyperparameters in the proposed model. Seed=12, dropout=0.4, network depth =4, learning rate= 1×10^{-2} , and batch size=64 are used. The SGD optimizer is used to optimise the model's weights. In our DenseNet implementation, we employed categorical cross entropy as a loss function. The model is optimised using the stochastic gradient descent approach, and the cost function for calculating the reconstruction loss is mean squared error. In order to avoid the model from being overfit, we adopted an early ending regularisation technique, which causes training to end when the loss stops improving.

4.2. Evaluation metrics

In the assessment procedure, the area under the curve (AUC) measurement used to describe the receiver operating characteristics is employed (ROC). The AUC demonstrates how differentiable two groups may be from one another, while the ROC is a probability curve. A highly good prediction model will have a large area under the curve (AUC), which shows that the model has a good measure of separability.[26]

FPR and TPR can be calculated as follows:

$$\text{TPR} = \frac{\sum TP}{\sum (TP + FN)}$$

$$\text{FPR} = \frac{\sum FP}{\sum (TP + FN)}$$

4.3. Comparative analysis on individual classes of UCF-Crime

Since the UCF-Crime dataset is the most current, only recently developed algorithms have reported their findings on it. Fig.5. shows the multiclass AUC curve with the AUC score of all the 14 classes present in the UCF-Crime dataset. We then look at the AUC performance on each individual anomaly class to see if our algorithm can detect small odd occurrences on the UCF-Crime dataset. From Fig.6. it can be inferred that our technique outperforms [6] [23] for all the abnormal classes except the two i.e., fighting and vandalism. From the Table 2 it can infer that the proposed model beats earlier state-of-the-art models.

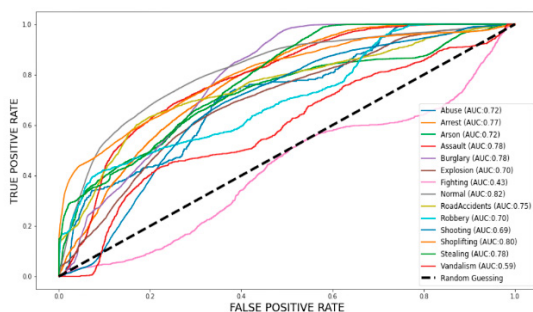


Fig.5. Multiclass AUC Curve

Table 2: Experimental Results

#	Method	Year	AUC%
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1	Waqas et al. with constraints [5]	2018	74.44
2	Waqas et al. no constraints [5]	2018	75.41
3	GODS [14]	2019	70.46
4	C3D [15]	2019	81.08
5	Spatiotemporal [24]	2019	63
6	GMM-based [25]	2020	75.90
7	MILR [16]	2020	76.67
8	Multiple instance-based [17]	2020	79.45
9	MIST [18]	2021	82.30
10	WSAL [19]	2021	85.38
11	Tian et al. [23]	2021	84.30
12	Proposed Model	-	86.61

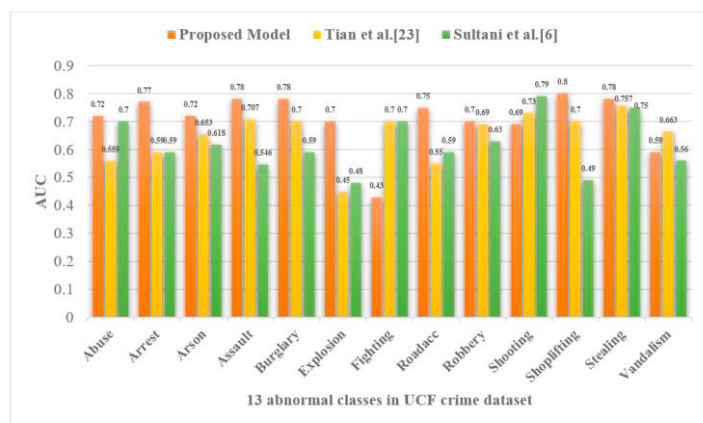


Fig.6. Results of the AUC on UCF-Crime for individual classes

5. Conclusion

This study proposes a method that can identify anomalous behavior in almost real-time. High Performance Computing through the usage of NVIDIA GPUs with creative solutions that may be used to monitor campuses and airports is one of the applications' advantages. The experimental findings show that the suggested technique successfully detects abnormal behavior in a promising manner that is almost as accurate as actual behavior. This research offered a unique computer vision method for detecting anomalous behavior such as robbery, assault, and other forms of criminal activity that can constitute a harm to the community. For the framework, we used a DenseNet like cascade of deep convolutional neural networks. After training a lightweight yet robust DenseNet, our system attained an overall accuracy of 86.63 percent. To increase the accuracy of the system, more supporting research is needed to eliminate complications such as shadows and non-leaving lookalikes engaged in abnormal behavior

identification. Future applications of the research, which may be used with satellite imagery, have a very broad reach. The project may be easily developed to track or research people's movements, which might be useful for keeping tabs on those participating in riots, demonstrations, etc. We also aim to test our approach with different sources of data.

References

- [1] Chandola, V., Banerjee, A., & Kumar, V. (2009). Anomaly detection: A survey. *ACM computing surveys (CSUR)*, 41(3), 1-58.
- [2] Ullah, H., Altamimi, A. B., Uzair, M., & Ullah, M. (2018). Anomalous entities detection and localization in pedestrian flows. *Neurocomputing*, 290, 74-86.
- [3] Kiran, B. R., Thomas, D. M., & Parakkal, R. (2018). An overview of deep learning based methods for unsupervised and semi-supervised anomaly detection in videos. *Journal of Imaging*, 4(2), 36.
- [4] Li, W., Mahadevan, V., & Vasconcelos, N. (2013). Anomaly detection and localization in crowded scenes. *IEEE transactions on pattern analysis and machine intelligence*, 36(1), 18-32.
- [5] Lu, C., Shi, J., & Jia, J. (2013). Abnormal event detection at 150 fps in matlab. In *Proceedings of the IEEE international conference on computer vision* (pp. 2720-2727).
- [6] Sultani, W., Chen, C., & Shah, M. (2018). Real-world anomaly detection in surveillance videos. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 6479-6488).
- [7] Calderara, S., Heinemann, U., Prati, A., Cucchiara, R., & Tishby, N. (2011). Detecting anomalies in people's trajectories using spectral graph analysis. *Computer Vision and Image Understanding*, 115(8), 1099-1111.
- [8] Jiang, F., Yuan, J., Tsafaris, S. A., & Katsaggelos, A. K. (2011). Anomalous video event detection using spatiotemporal context. *Computer Vision and Image Understanding*, 115(3), 323-333.
- [9] Tung, F., Zelek, J. S., & Clausi, D. A. (2011). Goal-based trajectory analysis for unusual behavior detection in intelligent surveillance. *Image and Vision Computing*, 29(4), 230-240.
- [10] Li, C., Han, Z., Ye, Q., & Jiao, J. (2013). Visual abnormal behavior detection based on trajectory sparse reconstruction analysis. *Neurocomputing*, 119, 94-100.
- [11] Zhang, Y., Lu, H., Zhang, L., & Ruan, X. (2016). Combining motion and appearance cues for anomaly detection. *Pattern Recognition*, 51, 443-452.
- [12] Wang, S., Zhu, E., Yin, J., & Porikli, F. (2018). Video anomaly detection and localization by local motion based joint video representation and OCELM. *Neurocomputing*, 277, 161-175.
- [13] Hasan, M., Choi, J., Neumann, J., Roy-Chowdhury, A. K., & Davis, L. S. (2016). Learning temporal regularity in video sequences. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 733-742).
- [14] Wang, J., & Cherian, A. (2019). Gods: Generalized one-class discriminative subspaces for anomaly detection. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 8201-8211).
- [15] Zhong, J. X., Li, N., Kong, W., Liu, S., Li, T. H., & Li, G. (2019). Graph convolutional label noise cleaner: Train a plug-and-play action classifier for anomaly detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1237-1246).
- [16] Dubey, S., Boragule, A., & Jeon, M. (2019, October). 3d resnet with ranking loss function for abnormal activity detection in videos. In *2019 International Conference on Control, Automation and Information Sciences (ICCAIS)* (pp. 1-6). IEEE.
- [17] Kamoona, A. M., Gosta, A. K., Bab-Hadiashar, A., & Hoseinnezhad, R. (2020). Multiple instance-based video anomaly detection using deep temporal encoding-decoding. *arXiv preprint arXiv:2007.01548*.
- [18] Feng, J. C., Hong, F. T., & Zheng, W. S. (2021). Mist: Multiple instance self-training framework for video anomaly detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 14009-14018).
- [19] Lv, H., Zhou, C., Cui, Z., Xu, C., Li, Y., & Yang, J. (2021). Localizing anomalies from weakly-labeled videos. *IEEE transactions on image processing*, 30, 4505-4515.
- [20] He, K., Zhang, X., Ren, S., & Sun, J. (2016). Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 770-778).

- [21] Huang, Gao, Zhuang Liu, Laurens Van Der Maaten, and Kilian Q. Weinberger. "Densely connected convolutional networks." In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pp. 4700-4708. 2017.
- [22] Szegedy, C., Liu, W., Jia, Y., Sermanet, P., Reed, S., Anguelov, D., ... & Rabinovich, A. (2015). Going deeper with convolutions. In *Proceedings of the IEEE conference on computer vision and pattern recognition* (pp. 1-9).
- [23] Tian, Y., Pang, G., Chen, Y., Singh, R., Verjans, J. W., & Carneiro, G. (2021). Weakly-supervised video anomaly detection with robust temporal feature magnitude learning. In *Proceedings of the IEEE/CVF International Conference on Computer Vision* (pp. 4975-4986).
- [24] Zhong, J. X., Li, N., Kong, W., Liu, S., Li, T. H., & Li, G. (2019). Graph convolutional label noise cleaner: Train a plug-and-play action classifier for anomaly detection. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition* (pp. 1237-1246).
- [25] Ul Amin, S., Ullah, M., Sajjad, M., Cheikh, F. A., Hijji, M., Hijji, A., & Muhammad, K. (2022). EADN: An Efficient Deep Learning Model for Anomaly Detection in Videos. *Mathematics*, 10(9), 1555.
- [26] Roka, S., Diwakar, M., & Karanwal, S. (2022). A Review in Anomalies Detection Using Deep Learning. In *Proceedings of Third International Conference on Sustainable Computing* (pp. 329-338). Springer, Singapore.